

Precalc Lesson 0.4 Homework

Answer
Key

1. $a = \frac{F}{m}$

2. $w = \frac{k - a + t}{3}$

3. $h = 6$ inches

4. D

5. The sphere; its volume is exactly $\frac{4}{3}$ times that of the cylinder.

6. The Aldi cantaloupe is bigger. Aldi cantaloupe has a radius of 8 inches; Meijer cantaloupe has a radius of 6 inches.

7. The 13-inch round baking pan. The volume of cake mix after baking is 175.5 cubic inches. Since the area of the base of the 9-in cake pan is only 63.63 square inches, the cake would end up being much taller (roughly 2.8 inches) which would affect the bake time. The 13-inch round baking pan would have a base area of 132.73 square inches, which is much closer to the original base surface area of $9 \times 13 = 117$ square inches. Using the 13-in round pan will produce a cake that is shorter than the original (roughly 1.32 inches tall) which is only about 0.18 inches shorter than the original cake.

8. C

9. $r = \sqrt{\frac{kQ_1Q_2}{F}}$

10. The force is 67% or roughly two-thirds of its original force.